TechXpress

E-commerce Platform with ASP.NET Core
Project Plan & Project Management

Full Stack .NET Project 1 | 2 Oct - 30 Nov 2026 | Team: 5 members

Start date	Friday, 2 October 2026	Delivery date	Monday, 30 November 2026
Project type	E-commerce web app	Architecture	N-layer + Repository + Unit of Work

1. Project Overview

Objective. Build a scalable, maintainable e-commerce platform for selling electronics (laptops, mobiles, cameras). Customers browse products, fill a cart and pay through Stripe; admins manage products, categories and orders. The code follows a layered architecture (Web, Services, Data), the Repository Pattern, Unit of Work and Dependency Injection. **The plan puts the weight on the .NET backend:** every member owns a backend slice from repository to controller.

1.1 Technology stack

Area	Technologies
Backend	ASP.NET Core MVC, Entity Framework Core, ASP.NET Identity, xUnit
Frontend	Razor views, Bootstrap, jQuery, DataTables, Toastr JS
Payments	Stripe (Checkout Session + Webhooks)
Hosting	Microsoft Azure (App Service + Azure SQL) or Hostinger
Collaboration	Git / GitHub, Trello or GitHub Projects

1.2 Scope

In scope	Out of scope (unless the team agrees)
• Product listing with search, filters, pagination • Registration, login, roles (Customer / Admin) • Session-based cart with discounts • Checkout and Stripe payment • Customer profile and order history • Admin dashboard (DataTables) • Deployment and documentation	• Mobile app • Product ratings and reviews • Multi-currency / multi-language • Courier shipping integration

1.3 Success criteria

- The full purchase flow works: browse, cart, checkout, payment confirmation.
- Role-based access is enforced: a customer can never reach an admin page.
- Code is split into three layers (Web, Services, Data) with no reference violations.
- Every member has delivered backend code (repositories, services, controllers) and tests for their slice.
- The site is deployed and reachable on a live URL by 30 November 2026.
- Documentation (architecture, setup guide, user manual) is complete.

2. Execution Phases and Calendar

The 60 days from 2 October to 30 November 2026 are split into four phases. Each phase ends with a milestone that the whole team tests together.

Phase	Dates	Focus	Key outputs	Milestone
Phase 1	2 - 15 Oct	Setup and product listing	3-layer solution, EF Core schema, repositories, Unit of Work, Identity, product services and listing page	M2 - 15 Oct
Phase 2	16 - 29 Oct	Cart, RBAC, admin panel	Session cart with discounts and stock checks, role-based access, admin CRUD with DataTables	M3 - 29 Oct
Phase 3	30 Oct - 12 Nov	Orders, Stripe, profile	Order placement in one transaction, Stripe Checkout + Webhook, profile, order history, admin orders	M4 - 12 Nov
Phase 4	13 - 30 Nov	Polish, testing, deployment	Responsive UI, security review, full testing, Azure deployment, final documentation	M5 - 30 Nov

Early milestone M1 (Monday 5 Oct): solution structure, entities and DbContext are on the `develop` branch so the whole team can build on them.

2.1 Weekly calendar

Week	Dates	Week	Dates
Week 1	2 - 8 Oct	Week 5	30 Oct - 5 Nov
Week 2	9 - 15 Oct (M2 on 15 Oct)	Week 6	6 - 12 Nov (M4 on 12 Nov)
Week 3	16 - 22 Oct	Week 7	13 - 19 Nov
Week 4	23 - 29 Oct (M3 on 29 Oct)	Week 8	20 - 26 Nov (code freeze 26 Nov)
		Final days	27 - 30 Nov (final testing, docs, delivery on 30 Nov)

3. Team Structure and Responsibilities

Member	Role	Main responsibilities
Mostafa Soliman Mohamed Salem	Tech Lead / Data Layer	Solution structure, entities, DbContext, all repositories, Unit of Work, Migrations, transactions, concurrency, Azure SQL
Ahmed Hassan Ali Abotayesh	Auth, Profile & Security	ASP.NET Identity, roles, RBAC, account / profile / user-role services, order history, security review
Doaa Ali Sayed Omar	Catalog & UI	Product, category and search services, catalog controllers, shared layout, live search, error handling, UI polish
Aya Tarek Maghawry Elhwary	Cart, Orders & Payment	Cart, order and payment services, checkout, Stripe Checkout Session, webhook with signature check and idempotency
Aya Kamel Soliman	Admin, QA & DevOps	Admin services and controllers, dashboard statistics, file upload, test plan, Azure deployment, documentation assembly

3.1 Backend ownership (.NET)

Every member owns a vertical slice: contracts, repository methods, services, controllers, views and xUnit tests for that slice.

Member	Data layer	Services layer	Web layer (controllers, middleware)
Mostafa	All repositories, UoW, Migrations, indexes, row version	Reviews contracts of every service	Program.cs, DI composition root, seed data
Ahmed	ApplicationUser mapping, user queries	AccountService, ProfileService, UserRoleService	Account, Profile, Admin users controllers; authorization filters
Doaa	Product query methods (with Mostafa)	ProductService, CategoryService, SearchService	Home, Product controllers, JSON search endpoint, exception middleware

Member	Data layer	Services layer	Web layer (controllers, middleware)
Aya Tarek	Order / payment query methods (with Mostafa)	CartService, OrderService, PaymentService (Stripe)	Cart, Checkout, Stripe webhook controllers
Aya Kamel	Dashboard aggregate queries (with Mostafa)	AdminProductService, AdminOrderService, DashboardService, FileService	Admin area controllers; Azure configuration

3.2 How every member takes part in every step

- **Design:** each member writes the interfaces for their own slice and agrees them with Mostafa before coding.
- **Build:** each member implements Data, Service, Controller and View code for their slice, in all four phases.
- **Test:** each member writes xUnit tests for their own services and joins every weekly joint test.
- **Review:** each member reviews one peer's Pull Requests (see 6.2).
- **Document:** each member writes the documentation section for their slice, with screenshots.
- **Deploy:** every member joins the deployed-site test on 26 - 27 Nov; Mostafa and Aya Kamel run the deployment.

3.3 Responsibility matrix (RACI)

R = does the work, A = accountable, C = consulted, I = informed.

Task	Mostafa	Ahmed	Doaa	Aya T.	Aya K.
Solution, entities, Migrations	R/A	I	I	C	I
Repositories and Unit of Work	R/A	I	C	C	C
Identity, roles, profile	C	R/A	I	I	C
Product, category, search services	C	I	R/A	I	I
Cart, orders, checkout	C	I	C	R/A	I
Stripe integration	C	I	I	R/A	C
Admin dashboard and services	C	C	C	C	R/A
Unit tests for own slice	R	R	R	R	R
Full system testing	C	C	C	C	R/A
Deployment to Azure	R	I	I	C	R/A
Final documentation	R	R	R	R	R/A

Golden rule: only Mostafa creates Migrations or changes table structure. Everyone else requests changes from him.

4. Timeline

4.1 Milestones

Milestone	Date	Completion condition
M1 - Core foundation	Mon 5 Oct	Solution, entities and DbContext on develop
M2 - End of phase 1	Thu 15 Oct	Repositories, UoW, seed data, Identity, product services and listing page
M3 - End of phase 2	Thu 29 Oct	Working cart, RBAC enforced, admin product and category management
M4 - End of phase 3	Thu 12 Nov	Orders with Stripe, profile, order history, admin orders
Code freeze	Thu 26 Nov	No new features; only bug fixes
M5 - Final delivery	Mon 30 Nov	Deployed site, full testing, complete documentation

4.2 Overview by member and phase

Member	Phase 1 (2-15 Oct)	Phase 2 (16-29 Oct)	Phase 3 (30 Oct-12 Nov)	Phase 4 (13-30 Nov)
Mostafa	Solution, entities, DbContext, repositories, UoW, seed data	OrderRepository, stock and discount fields, indexes, PR reviews	Transactions in UoW, payment fields, stock row version	Azure SQL, migrations in production, architecture docs
Ahmed	Identity, roles seed, AccountService, register / login	RBAC, AccessDenied, user-role service, anti-forgery	ProfileService, order history and details	Security review, auth and profile docs
Doaa	Layout, ProductService, CategoryService, listing page	Product details, home page, discount and stock display	Live search endpoint, AJAX filters, exception middleware	UI polish, catalog docs
Aya Tarek	Stripe study, cart and order contracts	CartService (Session), discounts, stock validation, cart page	OrderService, Stripe Checkout, webhook	Purchase-flow tests, live webhook test, payment docs
Aya Kamel	Admin layout, DataTables prototype, test project setup	Admin products and categories CRUD, FileService	Admin orders, dashboard service, test plan, Azure account	Full testing, Azure deployment, final docs assembly

4.3 Task breakdown by phase

Phase 1 - Setup and product listing (2 - 15 Oct)

Task	Owner	Dates	Depends on
Create solution and the three projects (Web, Services, Data), set references	Mostafa	2 - 3 Oct	None
Entities, DbContext, Fluent API, first Migration (M1)	Mostafa	2 - 5 Oct	Solution
Generic repository + Product / Category repositories	Mostafa	6 - 9 Oct	Entities
Unit of Work, DI registration, seed data	Mostafa	9 - 12 Oct	Repositories
Identity setup: ApplicationUser, roles seed, password policy	Ahmed	6 - 9 Oct	DbContext
AccountService, AccountController, register / login views	Ahmed	9 - 14 Oct	Identity setup
Main layout (Navbar, Footer, Bootstrap, jQuery, Toastr)	Doaa	2 - 5 Oct	None
ProductService and CategoryService: search, filters, sort, pagination	Doaa	6 - 13 Oct	Repositories
ProductController and listing page views	Doaa	10 - 15 Oct	ProductService
Study Stripe (Checkout, webhooks); test account and keys in User Secrets	Aya Tarek	2 - 8 Oct	None

Task	Owner	Dates	Depends on
Design ICartService, CartItem model, IOrderService contracts with Mostafa	Aya Tarek	9 - 14 Oct	Entities
Admin layout and server-side DataTables prototype	Aya Kamel	2 - 8 Oct	None
Admin area, IAdminProductService contract, xUnit test project setup	Aya Kamel	9 - 15 Oct	Services project
xUnit tests for own repositories and services	Each member	9 - 15 Oct	Own slice

Phase 2 - Cart, RBAC and admin panel (16 - 29 Oct)

Task	Owner	Dates	Depends on
OrderRepository, OrderDetail repository, stock and discount fields + Migration	Mostafa	16 - 20 Oct	Entities
Query optimization: indexes, AsNoTracking on read paths; PR reviews	Mostafa	21 - 29 Oct	Repositories
RBAC on admin controllers, AccessDenied page, role-based menu	Ahmed	16 - 21 Oct	Identity
User and role management service for admins	Ahmed	21 - 28 Oct	AccountService
Anti-forgery tokens on all POST actions	Ahmed	26 - 29 Oct	Forms
Product details page and home page (view models, related products query)	Doaa	16 - 23 Oct	ProductService
Discounted price display and stock status	Doaa	24 - 29 Oct	Discount fields
CartService (Session, JSON, CommitAsync): add, remove, update, clear, totals	Aya Tarek	16 - 22 Oct	ICartService
Discount logic and stock validation	Aya Tarek	22 - 26 Oct	CartService
CartController, cart page, Navbar counter	Aya Tarek	24 - 29 Oct	CartService
Admin products: DataTables endpoint, CRUD, image upload (FileService)	Aya Kamel	16 - 25 Oct	Product repository
Admin categories CRUD	Aya Kamel	25 - 29 Oct	Category repository
xUnit tests for own services; PR reviews and joint testing	All	26 - 29 Oct	Own slice

Phase 3 - Orders, Stripe and profile (30 Oct - 12 Nov)

Task	Owner	Dates	Depends on
Transaction support in Unit of Work, payment fields + Migration	Mostafa	30 Oct - 2 Nov	OrderRepository
Stock concurrency protection (row version) and integrity review	Mostafa	3 - 9 Nov	Stock fields
ProfileService and ProfileController: details, change password	Ahmed	30 Oct - 5 Nov	Identity
Order history and order details with ownership check	Ahmed	5 - 10 Nov	OrderService
Live search endpoint (JSON) and AJAX filter partial views	Doaa	30 Oct - 6 Nov	ProductService
Global exception handling middleware and logging; index review with Mostafa	Doaa	6 - 12 Nov	Indexes
Decide stock rule: reduce stock when payment is confirmed (with Mostafa)	Aya Tarek	30 Oct - 1 Nov	Transactions
Checkout page and OrderService (order and lines saved in one transaction)	Aya Tarek	30 Oct - 6 Nov	Transactions
Stripe Checkout Session, success and cancel pages	Aya Tarek	5 - 9 Nov	OrderService
Webhook: signature check, idempotent Pending to Paid update, stock reduction	Aya Tarek	8 - 12 Nov	Stripe session

Task	Owner	Dates	Depends on
Admin orders: list, details, status change	Aya Kamel	30 Oct - 6 Nov	OrderRepository
Dashboard statistics service and view; order filtering	Aya Kamel	6 - 12 Nov	Admin orders
Test plan (test cases) and Azure account setup	Aya Kamel	3 - 12 Nov	None
xUnit tests for own services; PR reviews and joint testing	All	11 - 12 Nov	Own slice

Phase 4 - Polish, testing and deployment (13 - 30 Nov)

Task	Owner	Dates	Depends on
Production connection string, Azure SQL, EnableRetryOnFailure, run migrations	Mostafa	13 - 22 Nov	Azure account
Architecture documentation (layers, Repository, UoW)	Mostafa	20 - 28 Nov	All code
Security review: authorization audit, ownership checks, cookies, HTTPS	Ahmed	13 - 21 Nov	All features
Authentication and profile documentation	Ahmed	22 - 28 Nov	Security review
UI polish: responsive layout, consistency, performance	Doaa	13 - 24 Nov	All pages
Catalog and UI documentation with screenshots	Doaa	22 - 28 Nov	UI polish
Purchase-flow and edge-case tests: empty cart, low stock, declined card, duplicate webhook	Aya Tarek	13 - 21 Nov	Stripe
Register webhook endpoint on the deployed site; live Test-mode payment	Aya Tarek	23 - 26 Nov	Deployment
Cart and payment documentation	Aya Tarek	22 - 28 Nov	Tests
Full system testing and bug logging	Aya Kamel	13 - 24 Nov	All features
Deploy to Azure (App Service, App Settings, HTTPS)	Aya Kamel	20 - 25 Nov	Azure SQL
Compile final documentation and user manual	Aya Kamel	25 - 29 Nov	Section drafts
Test deployed site (code freeze 26 Nov)	All	26 - 27 Nov	Deployment
Final review and delivery	All	29 - 30 Nov	Documentation

5. Deliverables

Date	Deliverable	Owners
5 Oct	Solution with three layers, entities, DbContext on develop	Mostafa
15 Oct	Layered setup, EF Core schema, repositories and Unit of Work, Identity, product services and listing	Mostafa, Ahmed, Doaa
29 Oct	Session cart, RBAC, admin products and categories, tests for each slice	Aya Tarek, Ahmed, Aya Kamel
12 Nov	Order placement with Stripe, profile and order history, admin orders and dashboard, tested payment flow	Aya Tarek, Ahmed, Aya Kamel
26 Nov	Code freeze; deployed site on Azure / Hostinger	Aya Kamel, Mostafa
30 Nov	Responsive UI, full testing report, final documentation and user manual	Everyone

5.1 Final delivery checklist

- ■ Solution with Web / Services / Data layers and clean references
- ■ EF Core database with repositories and Unit of Work
- ■ Authentication with Customer and Admin roles
- ■ Product page with search, filters and pagination
- ■ Working Session cart with discounts and stock checks
- ■ Order and lines saved in one transaction; stock reduced on confirmed Stripe payment via webhook
- ■ Customer profile and order history
- ■ Admin dashboard with DataTables for products, categories, orders
- ■ xUnit tests for every member's services
- ■ Responsive, consistent UI
- ■ Live deployment on Azure or Hostinger
- ■ Documentation: architecture explanation, setup guide, user manual
- ■ Organized GitHub repository with README

6. Project Management and Tracking

6.1 Git workflow

- **Branches:** `main` (stable, protected), `develop` (integration), `feature/<name>` (one per feature, e.g. feature/cart, feature/stripe).
- **Merging:** by Pull Request only, after review by another member.
- **Commits:** clear messages in one format, e.g. `feat: add cart service`.
- **Daily habit:** pull from develop before starting; merge latest develop into your branch before opening a PR.
- **Never commit:** Stripe keys, passwords, `bin` / `obj` folders (User Secrets locally, App Settings on Azure).

6.2 Peer code review

Code owner	Mostafa	Ahmed	Doaa	Aya Tarek	Aya Kamel
Reviewer	Aya Kamel	Mostafa	Ahmed	Mostafa	Doaa

Reviewers check: layer separation, no direct DbContext use in controllers, input validation, async/await use, clear naming, tests present.

6.3 Communication and meetings

Activity	Frequency	Duration	Purpose
Stand-up	Daily or every 2 days	10 min	Done, next, blockers
Weekly review	End of each week	45-60 min	Demo, joint testing, plan next week
Team chat (WhatsApp / Discord / Teams)	Ongoing	-	Quick questions and alerts

Activity	Frequency	Duration	Purpose
Task board (Trello / GitHub Projects)	Ongoing	-	Track task status

Rule: anyone blocked for more than half a day reports it immediately instead of waiting for the next meeting.

6.4 Task board and Definition of Done

Board columns: Backlog → To Do → In Progress → In Review → Testing → Done. Each card carries title, owner, phase, priority and acceptance criteria.

A task is **Done** only when:

- The project builds and the feature runs without errors.
- The owner tested the normal case and at least one error case, and wrote xUnit tests for new service logic.
- The Pull Request was reviewed and approved.
- It is merged into develop with no conflicts.
- The matching documentation section is written (short paragraph + screenshots).

6.5 Progress report (end of each phase)

Item	Phase 1	Phase 2	Phase 3	Phase 4
Planned tasks				
Completed tasks				
Completion %				
Blockers				
Decisions for next phase				

7. Quality and Testing

- **Individual testing:** before every Pull Request, with xUnit tests for new service logic.
- **Integration testing:** at the end of each week and each phase with everyone trying the merged build.
- **Full testing:** 13 - 24 Nov, led by Aya Kamel. Bugs are logged with steps to reproduce, screenshot, severity and owner; high-severity bugs are fixed before new features.

Area	Test scenario	Expected result
Registration	New email vs existing email	Success, then a clear error
Login	Correct and wrong credentials	Login, then rejection with message
Permissions	Customer opens an admin link	Redirect to AccessDenied
Products	Search + category + price filters together	Correct results with pagination
Cart	Quantity above stock	Rejected with a message
Cart	Change quantity, remove item	Total updates immediately
Order	Order placed, payment not yet confirmed	Order saved as Pending; stock unchanged
Order	Failure mid-process	Rollback, no partial save
Stripe	Successful test card	Webhook sets Paid and reduces stock once
Stripe	Same webhook delivered twice	Second delivery changes nothing
Stripe	Declined card or cancelled payment	Order stays unpaid; stock unchanged
Profile	Customer opens another customer's order	Access blocked
Admin	Add, edit, delete a product	Table updates immediately
Admin	Change order status	Status updates and shows to customer
Deployed site	Repeat full purchase flow	Same behavior as local

8. Risk Management

#	Risk	Likelihood	Impact	Mitigation
1	Foundation delayed, blocking others	Medium	High	Mostafa delivers by 5 Oct; others use mock data meanwhile
2	Migration conflicts	High	High	Only Mostafa creates Migrations
3	Git merge conflicts	High	Medium	Small branches, frequent merges, short PRs
4	Stripe / Webhook harder than expected	Medium	High	Aya Tarek studies from 2 Oct; Test mode; extra time in phase 3
5	Member absent or overloaded	Medium	High	Backup person per task; documented, reviewed code
6	Late Azure deployment problems	Medium	High	Azure account by 12 Nov; early trial deploy; deployment from 20 Nov
7	Leaked keys or connection strings	Low	High	User Secrets, Azure App Settings, review commits
8	Scope creep	Medium	Medium	Discuss in weekly meeting; add only after essentials
9	Inconsistent look between pages	Medium	Low	Shared layout and CSS from Doaa; review in phase 4
10	Not enough time for testing	High	High	xUnit from phase 1; test plan from 3 Nov; code freeze 26 Nov

Plan B if the project falls behind

Cut from the bottom of this priority list:

1. **Essential, never cut:** product browsing, cart, checkout, Stripe, Identity
2. Admin dashboard for products and orders
3. Profile and order history
4. Discounts and promotions
5. Live search and advanced UI polish
6. Statistics dashboard

Closing notes

- Two biggest success factors: the core foundation in the first days, and continuous testing instead of testing at the end.
- Following the Git and Migration rules avoids more than half of the expected problems.
- Any change to scope or dates is agreed in the team meeting and recorded.
- Each member documents their own part as they go.